# STUDENT ATTENDANCE RECORD SYSTEM

Project Report | Python 3 Console Application

## 1. INTRODUCTION

Student Attendance Record is a basic Python program to help students track their attendance in 4 subjects (CSE, EVS, English, Maths). It calculates the attendance percentage and checks if the student meets the 75% rule for CAT 1 and CAT 2 exams.

## 2. PROBLEM STATEMENT

Students often calculate attendance manually, which can cause calculation mistakes. Sometimes they do not realize their attendance is low until they cannot sit for exams. This simple tool helps them check eligibility quickly.

## 3. OBJECTIVES

- Take student name, subject, and class counts as input.
- Calculate attendance percentage accurately.
- Check if attendance is 75% or higher for CAT 1 and CAT 2.
- Stop invalid inputs from crashing the program.
- Allow checking another student record without restarting.

## 4. FUNCTIONAL REQUIREMENTS

- Accept student name (cannot be left blank).
- Restrict subjects to CSE, EVS, English, or Maths.
- Accept total classes (> 0) and attended classes (>= 0).
- Calculate percentage using: (Attended / Total) * 100.
- Show 'ELIGIBLE' if >= 75%, else show 'NOT ELIGIBLE'.
- Prompt the user to check another record or exit.

## 5. NON-FUNCTIONAL REQUIREMENTS

- Simple Code: Clean, compact, and around 50-60 lines.
- Fast Execution: Runs instantly in the terminal.
- Easy Setup: Works on any system with standard Python 3.
- Clear Output: Displays a neat, aligned report card.

## 6. TECHNOLOGY & IMPLEMENTATION

Written in standard Python 3 with zero external libraries:
- input() and print() for terminal user interaction.
- while loops to re-prompt whenever inputs are invalid.
- if-else conditions to verify the 75% exam criterion.
- Functions: get_valid_subject(), get_positive_int(), and check_attendance().

## 7. WORKFLOW

1. Start program in terminal.
2. Enter student name.
3. Choose subject (CSE, EVS, English, or Maths).
4. Enter total classes and attended classes.
5. Program calculates percentage and checks 75% rule.
6. Displays attendance report with eligibility status.
7. Enter 'yes' to check another record or 'no' to exit.

## 8. TESTING

Tested with various input scenarios:
- Eligible: 32/40 classes = 80.00% -> Shows ELIGIBLE.
- Shortage: 25/40 classes = 62.50% -> Shows NOT ELIGIBLE.
- Exactly 75%: 30/40 classes = 75.00% -> Shows ELIGIBLE.
- Invalid Subject: Entering 'Physics' shows error and asks again.
- Invalid Numbers: Attended > total shows error and asks again.

## 9. RESULT

The program executes smoothly in the terminal without errors. It provides an accurate attendance percentage, clearly states exam eligibility for CAT 1 and CAT 2, and handles invalid inputs reliably.

## 10. CHALLENGES

- Keeping code short (50-60 lines) while including thorough validation.
- Preventing zero division error when total classes is entered as zero.
- Supporting subject names in both small and capital letters (e.g., cse or CSE).

## 11. LEARNINGS

- How to structure code into small, clean functions.
- How to use while loops for safe input validation.
- How to evaluate rules using conditional logic (if-else).
- How to format numbers neatly with two decimal places (:.2f).

## 12. FUTURE SCOPE

- Allow adding custom subjects instead of only 4 fixed ones.
- Save attendance records to a simple text or CSV file.
- Allow checking multiple subjects at once to calculate overall attendance.

## 13. REFERENCES


- Python Official Website: https://www.python.org
- Python Documentation: https://docs.python.org